

3. PROJECT DESIGN PHASE

3.1 System Design

The Incident Lifecycle Management solution runs as a single, centralized workflow inside ServiceNow.

At a high level, the flow moves as follows:

Incident Reported ↓ **Ticket Logged** ↓ **Incident Classified** ↓ **Impact and Urgency Assessed** ↓ **Priority Calculated** ↓ **Support Group Assigned** ↓ **First-Line Investigation** ↓ **Knowledge Base Searched** ↓ **Resolved, or Escalated** ↓ **Level 2 Review** ↓ **Child Incident / Emergency Change Raised if Needed** ↓ **Fix Identified** ↓ **Incident Resolved** ↓ **Knowledge Article Updated or Created** ↓ **SLA and Related Records Checked** ↓ **Incident Closed**

3.2 Designing the Incident Lifecycle

Stage 1 — New

An incident record is opened as soon as a user reports a problem, capturing the initial details.

Stage 2 — Classification

Category, subcategory, impact, and urgency are set, and priority is derived from how severe the issue is.

Stage 3 — Assignment

The ticket is routed to the support group or technical team best suited to handle it.

Stage 4 — Investigation

The assigned agent looks into the issue and checks the knowledge base for a known fix.

Stage 5 — Escalation

Where first-line support can't close the ticket, it moves up to Level 2 support.

Stage 6 — Related Activities

Depending on what the investigation uncovers, the team may:

- Open a child incident.
- Attach an existing knowledge article.
- Raise an emergency change request.
- Add further investigation notes.
- Reassign the ticket to another group.

Stage 7 — Resolution

Once the underlying issue is fixed, the incident is marked resolved with the appropriate resolution code and notes.

Stage 8 — Closure

After the fix is confirmed and related records are checked, the incident is closed out.

3.3 Data Design

The core record in this design is the **Incident** record, with these key fields:

- Incident Number
- Caller
- Service
- Service Offering
- Configuration Item
- Category
- Subcategory
- Short Description
- Description
- State
- Impact
- Urgency
- Priority
- Assignment Group
- Assigned To
- Resolution Code
- Resolution Notes

3.4 Integration Design

Incident ↔ Knowledge

Agents consult the knowledge base while investigating; existing articles can supply a ready-made fix.

Incident ↔ Change

Where an urgent system change is needed to fix an incident, an emergency change record is created and linked to it.

Parent ↔ Child Incidents

Related tickets are grouped through parent-child links, simplifying handling of a shared underlying issue.

Incident ↔ SLA

Each incident carries SLA data used to measure response and resolution performance.

3.5 Role Design

Interaction across the system is organized by role:

End User → raises the problem **Service Desk Agent** → logs, classifies, investigates, and updates the incident **Level 2 Support** → carries out advanced troubleshooting **Change Team** → manages emergency changes **Knowledge Team** → curates the knowledge base **Administrator** → configures the ServiceNow environment